Artin - linearly independence of characteristic theorem.

Def. Let G be a group and L be a field.
A group-Homomorphism $\chi_i : G \to L^\times$ is called a character of G in L.
The characters $\chi_1, \cdots, \chi_n$ are said to be linearly independent over L if
$\sum_{i=1}^{n} a_i \cdot \chi_i \equiv 0$ for any $a_i \in L$ implies $a_1 = \cdots = a_n = 0$.

Thm. If $\chi_1, \cdots, \chi_n : G \to L^\times$ are distinct characters of G in L,
then they are linearly independent over L.

Proof. Suppose that $\chi_1, \cdots, \chi_n$ are linearly dependent.
Choose $J \subseteq [n]$ with minimum cardinality among all non-empty sets satisfying
$\sum_{k \in J} a_k \cdot \chi_k \equiv 0$ and $\forall k \in J,\ a_k \neq 0$. $\cdots$ (*), with the coefficients a_i.
Since the characters are dependent, there are subsets of $[n]$ satisfying such conditions, so we can take this J. Moreover, since each Charac-ters have only non-zero valued, so J must have at least two elements.
Let $i, j \in J$ with $i \neq j$. Since $\chi_i \neq \chi_j$, for some $g_0 \in G$, we have

$$\chi_i(g_0) \neq \chi_j(g_0)$$

Given $g \in G$, $0 = \sum_{k \in J} a_k \cdot \chi_k(g_0 g) = \sum_{k \in J} a_k \cdot \chi_k(g_0) \cdot \chi_k(g)$. And, multiplying $\chi_j(g_0)$ to (*), $0 = \sum_{k \in J} a_k \cdot \chi_j(g_0) \cdot \chi_k(g)$. Now, substracting the two equations,
then we have $\sum_{k \in J} a_k \cdot (\chi_k(g_0) - \chi_j(g_0)) \cdot \chi_k(g) = 0$.
But, the coefficient of $\chi_j(g)$ is zero and of $\chi_i(g)$ is not zero,
so it is also non-trivial representation with the non-empty index set $J \subsetneq I$,
it contradicts to (*). Consequently, the chacters are independent. □

Note: $\mathcal{P} := \left\{ (a_1, \cdots, a_n) \in L^n \setminus \{(0, \cdots, 0)\} \,\middle|\, \sum_{k \in [n]} a_k \cdot \chi_k \equiv 0 \right\}$ is non-empty iff
$\{\chi_1, \cdots, \chi_n\}$ is linearly dependent.

View of Representation theory

Let $F^G := \{f \mid f: G \to F\}$ where G is a group and F is a field.

Then, F^G is a vector space under the operations

$$(f_1 + f_2)(g) := f_1(g) + f_2(g) \quad \text{and} \quad (af)(g) := a \cdot f(g).$$

Let $h \in G$ be given. Define

$$T_h : F^G \to F^G : f \mapsto T_h f \quad \text{where} \quad T_h f : G \to F : g \mapsto f(hg).$$

Normal basis theorem.

Def. Let E/F be a Galois extension field of degree of n.
For fixed $\gamma \in E$, a set

$$\eta = \{\sigma_1(\gamma), \cdots, \sigma_n(\gamma)\} \quad \text{where } Gal(E/F) = \{\sigma_1, \cdots, \sigma_n\}.$$

is called a normal basis for E/F if η is a basis.

Lemma. Let E/F be Galois extension with $Gal(E/F) = \{\sigma_1, \cdots, \sigma_n\}$.
For a basis $\beta = \{e_1, \cdots, e_n\}$ for E over F,

$$A = \{(\sigma_1(e_i), \cdots, \sigma_n(e_i)) \mid 1 \le i \le n\}$$

forms an E-basis for E^n

Proof. Let $W = Span_E(A) \le E^n$. Suppose that $W \neq E^n$.
Then, there exists a nonzero linear functional $\varphi : E^n \to E$ which vanishes on W. (In other words, $W \neq E^n$ implies $W^\circ \neq (E^n)^\circ = \{0\}$.)
Let $\{f_1, \cdots, f_n\}$ be the dual basis for $(E^n)^*$ corresponding to the standard basis $\{\alpha_1, \cdots, \alpha_n\}$ for E^n. For each $1 \le i \le n$,

$$\varphi\left((\sigma_j(e_i))_{j=1}^n\right) = \sum_{j=1}^n \varphi(\alpha_j) \cdot \sigma_j(e_i) = 0, \text{ since } \varphi \in W^\circ.$$

For any $x \in E$, x is F-linear combination by β, so that

$$x = c_1 e_1 + \cdots + c_n e_n \text{ for some } c_i \in F$$

Now, Since each σ_i's are field automorphism which F fixes, so

$$\begin{aligned}\sum_{j=1}^n \varphi(\alpha_j) \cdot \sigma_j(x) &= \sum_{j=1}^n \varphi(\alpha_j) \cdot \sigma_j\left(\sum_{i=1}^n c_i e_i\right) \\ &= \sum_{j=1}^n \varphi(\alpha_j) \cdot \sum_{i=1}^n c_i \cdot \sigma_j(e_i) \\ &= \sum_{i=1}^n c_i \cdot \sum_{j=1}^n \varphi(\alpha_j) \cdot \sigma_j(e_i) = \sum_{i=1}^n c_i \cdot 0 = 0.\end{aligned}$$

Now, by the independentness of the characters $\sigma_1|_{E^\times}, \cdots, \sigma_n|_{E^\times}$,

$$\varphi(\alpha_1) = \cdots = \varphi(\alpha_n) = 0.$$

But, it implies $\varphi \equiv 0$, so contradiction. □

Thm. Every finite Galois extension of an infinite field has a normal basis

Proof. Let E/F be the finite Galois extension. Denote

$$Gal(E/F)=\{\sigma_1,\cdots,\sigma_n\} \text{ where } n=[E:F], \text{ and } \sigma_1=id.$$

Given $x\in E$, consider F-linear relation

$$\sum_{j=1}^{n} a_j\cdot\sigma_j(x)=0 \quad \text{for some } a_1,\cdots,a_n\in F.$$

For each $1\le i\le n$,

$$\sigma_i^{-1}\left(\sum_{j=1}^{n} a_j\cdot\sigma_j(x)\right)=\sum_{j=1}^{n} a_j\cdot(\sigma_i^{-1}\circ\sigma_j)(x)=\sigma_i^{-1}(0)=0, \text{ so that}$$

$$\begin{pmatrix}(\sigma_1^{-1}\circ\sigma_1)(x) & \cdots & (\sigma_1^{-1}\circ\sigma_n)(x)\\ \vdots & \ddots & \\ (\sigma_n^{-1}\circ\sigma_1)(x) & \cdots & (\sigma_n^{-1}\circ\sigma_n)(x)\end{pmatrix}\cdot\begin{pmatrix}a_1\\ a_2\\ \vdots\\ a_n\end{pmatrix}=\begin{pmatrix}0\\ 0\\ \vdots\\ 0\end{pmatrix} \quad \cdots (*)$$

Therefore, the linear relation above is trivial if and only if the equation $(*)$ has only trivial solution if and only if the matrix in $(*)$ has nonzero determinant.

Let $\beta=\{\alpha_1,\cdots,\alpha_n\}$ be a basis for E over F.

And, write an arbitrary $x\in E$ as $x=\sum_{k=1}^{n} a_k\cdot\alpha_k$ where $a_k\in F$. Then, for each $1\le i,j\le n$, $(\sigma_i^{-1}\circ\sigma_j)(x)=\sum_{k=1}^{n} a_k\cdot(\sigma_i^{-1}\circ\sigma_j)(\alpha_k)$.

Consider the polynomial on variables $X_1,\cdots,X_n$ over E:

$$\Delta(X_1,\cdots,X_n)\overset{\text{def}}{=}\det\left(\sum_{k=1}^{n}(\sigma_i^{-1}\circ\sigma_j)(\alpha_k)\cdot X_k\right)_{\substack{1\le i\le n\\ 1\le j\le n}}$$

Proof, Page 2.

Meanwhile, observe that: Consider a matrix

$$\left(\sigma_i(\alpha_j)\right)_{i,j} \in M_{n\times n}(E)$$

Then, it is invertible, because: assume that this matrix is not invertible. Then, the rows are linearly dependent, so that there are not all zero scalars $a_1, \cdots, a_n \in E$ s.t

$$\text{for all } 1\le k\le n,\ \sum_{i=1}^{n} a_i \cdot \sigma_i(\alpha_k) = 0.$$

Now, let $x \in E$ be given. Denote $x = \sum_{k=1}^{n} b_k \alpha_k$ where $b_k \in F$.

Then, $\sum_{i=1}^{n} a_i \sigma_i(x) = \sum_{i=1}^{n} a_i \cdot \sigma_i\left(\sum_{k=1}^{n} b_k \cdot \alpha_k\right) = \sum_{k=1}^{n} b_k \cdot \sum_{i=1}^{n} a_i \cdot \sigma_i(\alpha_k) = 0.$

But, it contradicts to the independence of characters.

Now, the equation

$$\begin{pmatrix} \sigma_1(\alpha_1) & \cdots & \sigma_1(\alpha_n) \\ \vdots & \ddots & \vdots \\ \sigma_n(\alpha_1) & \cdots & \sigma_n(\alpha_n) \end{pmatrix} \cdot \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

has a unique solution, $(c_1, \cdots, c_n) \in E^n$. Since $\sigma_1 = id$, so that

$$\sum_{k=1}^{n} c_k \cdot \alpha_k = 1 \quad \text{and} \quad \sum_{k=1}^{n} c_k \cdot \sigma_j(\alpha_k) \quad (2 \le j \le n).$$

That is, $\sum_{k=1}^{n} c_k \cdot (\sigma_i^{-1} \circ \sigma_j)(\alpha_k) = \delta_{i,j}$. Therefore, $\Delta(c_1, \cdots, c_n) = \det I_n = 1$. So Δ is non-zero polynomial.

So, the non-zero polynomial Δ has only finite number roots and since F is infinite, there are $b_1, \cdots, b_n \in F$ such that $\Delta(b_1, \cdots, b_n) \neq 0$.

Now, $0 \neq \Delta(b_1, \cdots, b_n) = \det\left(\sum_{k=1}^{n} (\sigma_i^{-1} \circ \sigma_j)(\alpha_k) \cdot b_k\right) = \det\left((\sigma_i^{-1} \circ \sigma_j)(x)\right)$

where $x = \sum_{k=1}^{n} b_k \cdot \alpha_k$, so

$$\{\sigma_1(x), \cdots, \sigma_n(x)\} \text{ is a normal basis for } E \text{ over } F. \square$$